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SARDAR PATEL UNIVERSITY
B.C.A (5th SEM) - EXAMINATION 2026
US05DBCA56 - COMPUTER GRAPHICS



Date: 12-2-2026, Thursday

Time: 10:00 AM to 01:00 PM

NOTE: Right hand figure indicates the marks of each question.

Total Marks: [70]

Q. 1 Multiple Choice Questions (One marks each)**[10]**

- 1 CAD stands for _____.
 (A) Computer Aided Design (B) Common Application Design
 (C) Common Aided Design (D) Computer Application Design
- 2 _____ is not application of computer graphics.
 (A) Computer Art (B) Power point
 (C) Presentation Graphics (D) CAD
- 3 CRT stands for _____.
 (A) Cathode Ray Tube (B) Computer Ray Tube
 (C) Common Ray Tube (D) None of the above
- 4 DDA stands for _____.
 (A) Data Drawing Algorithm (B) Digital Drawing Algorithm
 (C) Digital Differential Analyzer (D) Digital Drawing Analyzer
- 5 _____ referred to a set of cast metal character forms.
 (A) Font (B) Character
 (C) Metal Words (D) Casted Words
- 6 _____ rule (s) used to identify interior region of an object.
 (A) ODD – EVEN Rule (B) Non Zero Winding no. rule
 (C) Polygon fill (D) Both (A) & (B)
- 7 Rotation Point is also known as _____.
 (A) Rotation Angle (B) Rotation Axis
 (C) Pivot Point (D) Translation Vector
- 8 A _____ is a transformation that produces a mirror image of an object.
 (A) Translation (B) Reflection
 (C) Rotation (D) Shear
- 9 _____ Mode application initiates data entry.
 (A) Request Mode (B) Event Mode
 (C) Sample Mode (D) None of above
- 10 3D Graphics include _____.
 (A) Height (B) Depth
 (C) Width (D) All of the above

Q. 2 Short Questions. (Attempt any ten)

[20]

- 1 What are disadvantages of Beam Penetration Method?
- 2 State disadvantages of DVST (Direct View Storage Tube).
- 3 Explain the terms (1) Bitmap, (2) Pixel.
- 4 What is inside-outside test? List various methods used for it.
- 5 List caps used at the end of line. Explain any one.
- 6 Give difference between Sans Serif Font and Serif Font.
- 7 What is clipping? List out all clipping object.
- 8 What is Window to Viewport transformation?
- 9 What is scaling? List out its types.
- 10 Define: Application icons and control icons.
- 11 List various input modes.
- 12 Explain in brief virtual reality environment.

Q. 3 A] Differentiate between Raster scan displays and Random scan displays.

[05]

B] Write note on DVST.

[05]

OR

Q. 3 A] Explain the working of CRT in detail.

[05]

B] List all types of input devices and explain any two in detail.

[05]

Q. 4 A] Describe DDA algorithm with example.

[05]

B] Write a short note on character generation.

[05]

OR

Q. 4 A] What is inside—outside test? Explain odd-even rule with example.

[05]

B] Explain Bresenham's line drawing algorithm with example.

[05]

Q. 5 A] Write a short note on Viewing Pipeline.

[05]

B] Explain in brief Cohen Sutherland Line Clipping Algorithm.

[05]

OR

Q. 5 A] What is geometric transformation? List all 2-D geometric transformation and explain any one in detail.

[05]

[05]

B] Explain in brief Sutherland Hodgeman polygon clipping algorithm.

Q. 6 A] Explain purpose of Graphical input function and different input modes.

[05]

B] List and explain Interactive picture-construction techniques

[05]

OR

Q. 6 A] List and explain Logical classification of Input Devices in detail.

[05]

B] Explain in detail General considerations in structuring a user dialogue.

[05]
